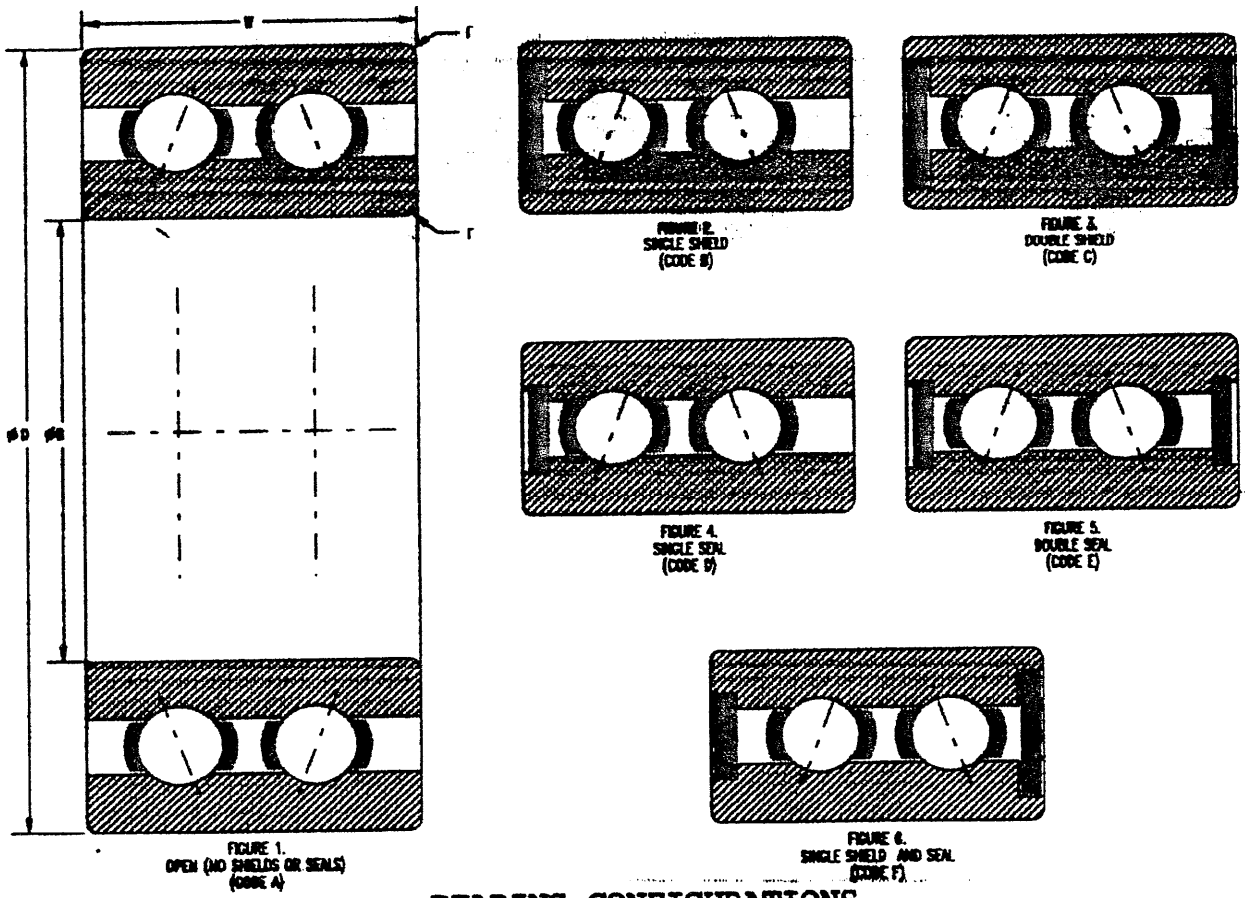
METRIC
FF-B-171/33
30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, DOUBLE ROW, ANGULAR CONTACT, NON-FILLING SLOT, VERTEX OF CONTACT ANGLES OUTSIDE BEARING, ANSI/AFBMA DIMENSION SERIES 33.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.



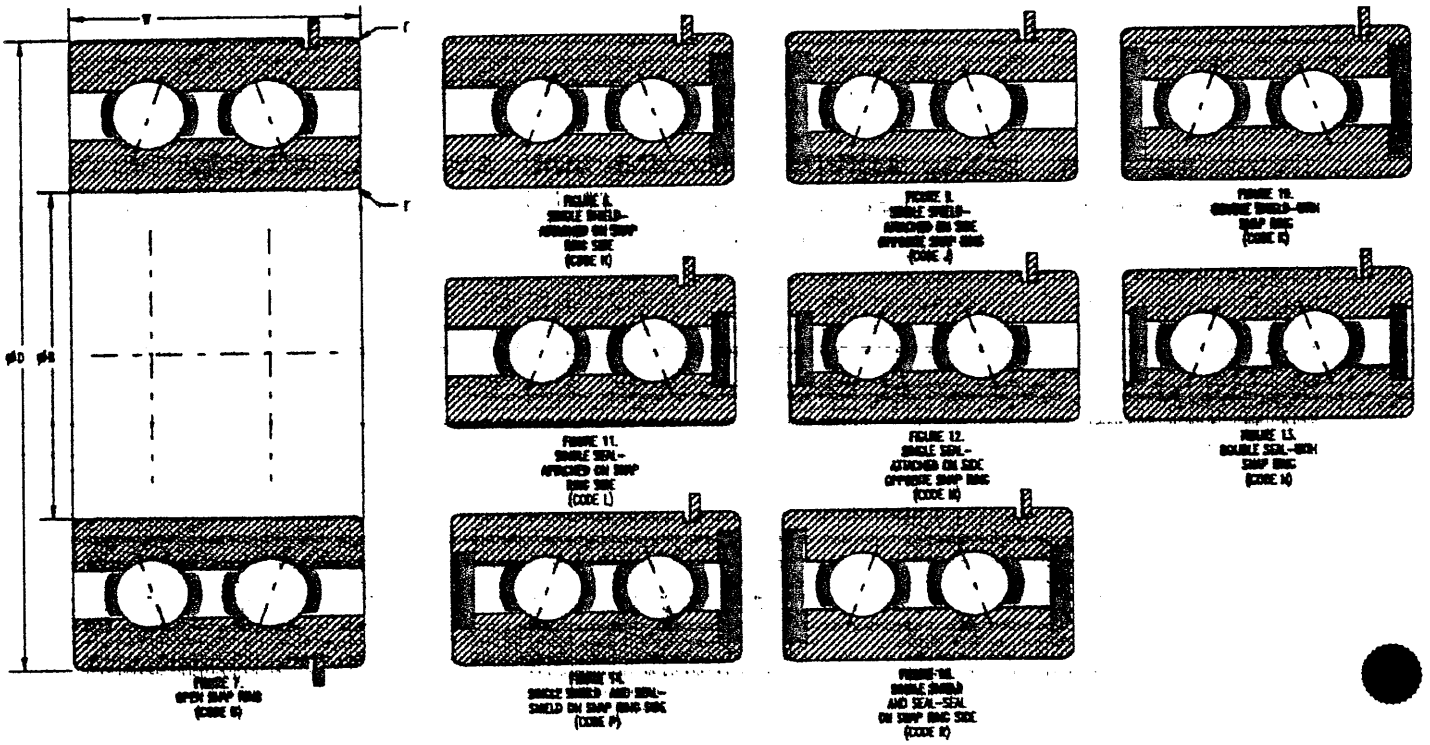
BEARING CONFIGURATIONS

AMSC/NA

FSC 3110

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.



BEARING CONFIGURATIONS - (Continued)

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:46Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. **Boundary dimensions.** The boundary dimensions shall be specified in the part number by the size codes in Table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. **Shields, seals, and snap rings.** Shields and seals shall be fabricated from any material unless otherwise specified and shall be removable or nonremovable type. Seals shall be impervious contact type. Snap rings shall be fabricated from ferrous material unless otherwise specified and shall have sufficient spring temper to set firmly in the ring grooves and shall be removable without the metal taking a permanent set. The snap ring and snap ring groove dimensions and their relative locations shall be in accordance with ANSI/AFBMA Standard 20. The number and positioning of shields, seals, and snap rings required shall be specified in the part number by the codes in table III.

4. **Radial Internal Clearance.** The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

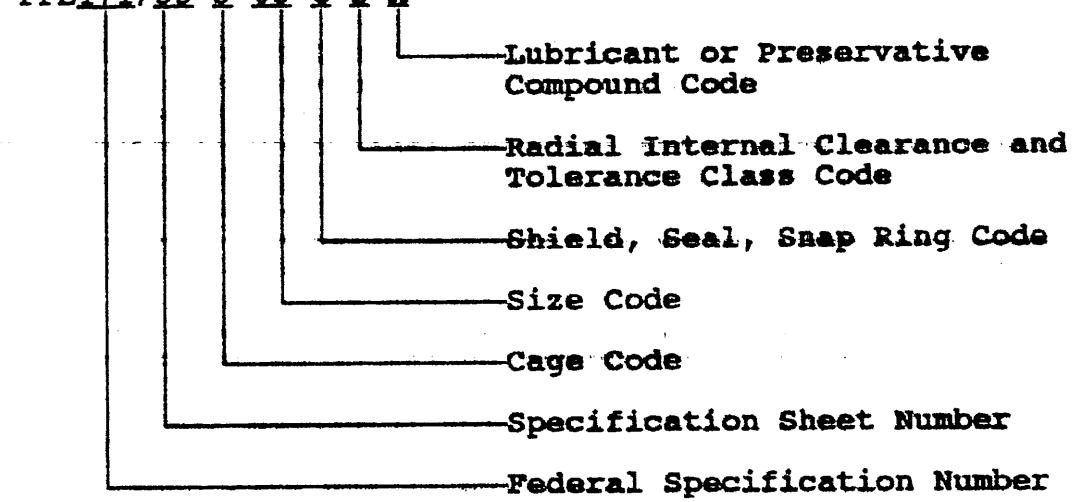
5. **Tolerance Class.** The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

6. Lubricant or Preservative Compound. Preservative compound shall be applied to all open bearings (no shields or seals) and to single closure bearings (single shield or single seal) that have no additional lubricant requirement. Thus, for these bearings the bearing part number shall include the “C” suffix. For double closure or single closure bearings that require lubricant the bearing part number shall include the “A” or “B” suffix l When grease is required, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25 to 50 percent filled. The lubricant or preservative compound shall be specified in the part number by the codes in table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.

7. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and-alpha or numeric codes for cage, size, shields, seals, and snap rings, radial internal clearance and tolerance class, and lubricant or preservative compound.

EXAMPLE: FFB171/33 J 03 G 2 A



FFB171/33J03G2A indicates - Bearing, cage-pressed steel or molded plastic: bore 17mm; outside diameter 4.7mm; width 22.2mm; open (no shields or seals) with snap ring; radial internal clearance symbol 0 (normal); tolerance class ABEC-1; grease in accordance with MIL-G-81322.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.

TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic load rating (pounds)
00	10	35	19.0	0.6	15	1730
01	12	37	19.0	1.0	18	1960
02	15	42	19.0	1.0	21	2170
03	17	47	22.2	1.0	23	3220
04	20	52	22.2	1.1	27	3725
05	25	62	25.4	1.1	32	5040
06	30	72	30.2	1.1	37	6910
07	35	80	34.9	1.5	43.5	8170
08	40	90	36.5	1.5	48.5	9985
09	45	100	39.7	1.5	53.5	11960
10	50	110	44.4	2.0	60	14090
11	55	120	49.2	2.0	65	17550
12	60	130	54.0	2.1	72	20060
13	65	140	58.7	2.1	77	22720
14	70	150	63.5	2.1	82	25525
15	75	160	68.3	2.1	87	27790
16	80	170	68.3	2.1	92	30105
17	85	180	73.0	3.0	99	32475
18	90	190	73.0	3.0	104	34895
19	95	200	77.8	3.0	109	38600

1/ Size codes for Table I are industry bore code numbers.

2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fine? radius. This dimension does not control the shape) of the bearing chamfer contour.

3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:46Z
Check the source to verify that this is the current version before use.

TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)
Y	Pressed brass

TABLE III. Codes for Shields and Seals.

Code	Shield, seal, and snap ring configuration
A	open (no shields or seals) - fig.1
B	single shield - fig.2
C	double shield - fig.3
D	single seal - fig.4
E	double seal - fig.5
F	single shield and seal - fig.6
G	open with snap ring - fig.7
H	single shield (attached on snap ring side) - fig.8
J	single shield (attached on side opposite snap ring)-fig.9
K	double shield with snap ring - fig.10
L	single seal (attached on snap ring side) - fig.11
M	single seal (attached on side opposite snap ring)-fig.12
N	double seal with snap ring - fig.13
P	single shield and seal (shield on snap ring side)-fig.14
R	single shield and seal (seal on snap ring side) - fig.15

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (less than normal)	ABEC-1
2	Symbol 0 (normal)	
3	Symbol 3 (greater than normal)	
4	Symbol 4 (greater than symbol 3)	
5	Symbol 2 (less than normal)	ABEC-5
6	Symbol 0 (normal)	
7	Symbol 3 (greater than normal)	
8	Symbol 4 (greater than symbol 3)	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.

TABLE V. Codes for Lubricant and Preservative Compound.

Code	Lubricant or Preservative Compound
A	Grease in accordance with MIL-G-81322
B	Grease in accordance with DOD-G-24508
C	Preservative compound in accordance with MIL-P-197

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:

PREPARING ACTIVITY:
DLA-IS

(Project 3110-0848-33)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.